Car Plate Recognition and Reconstruction with Deep Learning

Computer Vision – A.Y. 2024-2025

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Outline

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Introduction

Our goal was to develop a computer vision system to **detect license plates in images** using bounding box localization and extract text through optical character recognition, handling varied lighting and viewing angles for real-world accuracy.

1

Baseline Development

Create proprietary baseline model

2

Model Implementation

Develop Yolov5 and PDLPR models

3

Testing & Evaluation

Benchmark all models

4

Performance Comparison

Compare results and document findings

Yolov5-PDLPR Architecture Overview

Object Detection: **Yolov5**

IGFE

Encoder

Parallel Decoder

PDLPR

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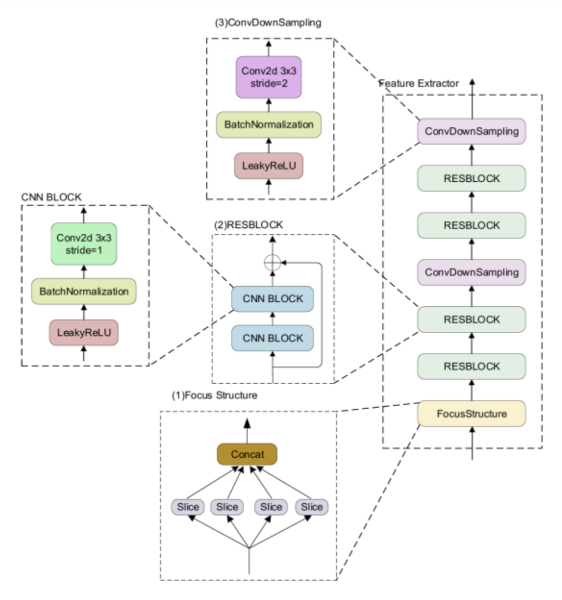
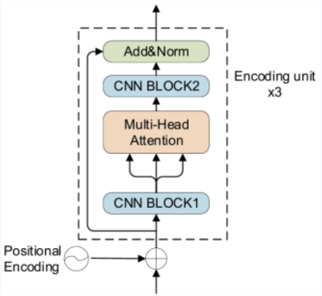
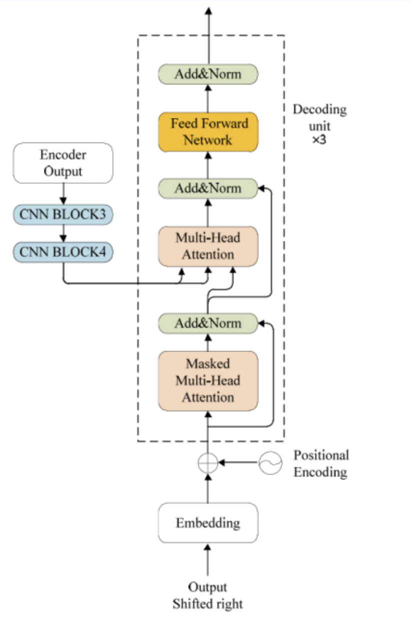
How PDLPR works



IGFE



Encoder



Decoder

ResNet18 Architecture Overview

Same Convolutional Architecture: **ResNet18**

1

Bounding Box linear part

2

OCR linear part

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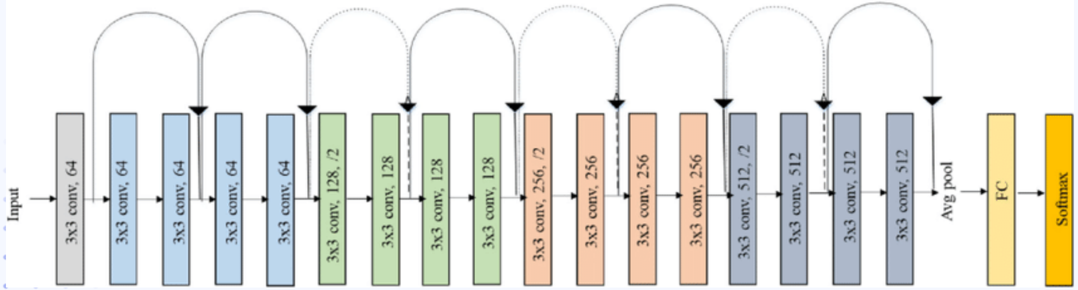


CCPD (Chinese City Parking Dataset)

A large-scale, publicly available dataset for license plate detection and recognition. Contains over **250,000** images of vehicles with annotated license plates.

Dataset Characteristics:
- Diverse conditions: various lighting, weather, and viewpoints.
- Plates captured in real-world scenarios including occlusion, blur, and distortion.
- Annotations include bounding boxes for plates and character-level labels.

Examples of possible car plates

fn

db

tilt

rotate

blur

weather

The Experimental setup (1)

Yolov5

- **Loss function**: GIoU loss
- **Optimizer**: Adam, with LR: 0.001
- **Batch**: 50
- **Epochs**: 15

$$L_{IoU} = 1 - IoU\left(B_{Pred}, B_{gt}\right)$$

$$IoU(A,B) = \frac{A \cap B}{A \cup B}$$

PDLPR

- **Loss function**: Cross Entropy
- **Optimizer**: Adam, with LR: $1e^{-4}$
- **Batch**: 128
- **Epochs**: 50

$$L_{OCR} = -\frac{1}{T}\sum_{i=1}^{T}\sum_{k=1}^{C} y_{i,k}\log(\hat{y}_{i,k})$$

The Experimental setup (2)

RN18 Bounding Box

- **Loss function**: Mean Squared Error
- **Optimizer**: Adam, with LR: $1e^{-3}$
- **Batch**: 50
- **Epochs**: 30

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

RN18 OCR

- **Loss function**: Cross Entropy
- **Optimizer**: Adam, with LR: $1e^{-4}$, weight decay: $5e^{-5}$
- **Batch**: 128
- **Epochs**: 30

$$L_{OCR} = -\frac{1}{T}\sum_{i=1}^{T}\sum_{k=1}^{C} y_{i,k}\log(\hat{y}_{i,k})$$

Yolov5 and RN18 BB results

IoU > 0.70

dataset	Yolov5	RN18 OCR
base	96,8%	88.7%
blur	89,4%	80.1%
challenge	87,7%	82.1%
db	78,3%	76.2%
fn	78,4%	75.8%
rotate	89,2%	84.3%
tilt	87,2%	82.4%
weather	92,5%	86.2%

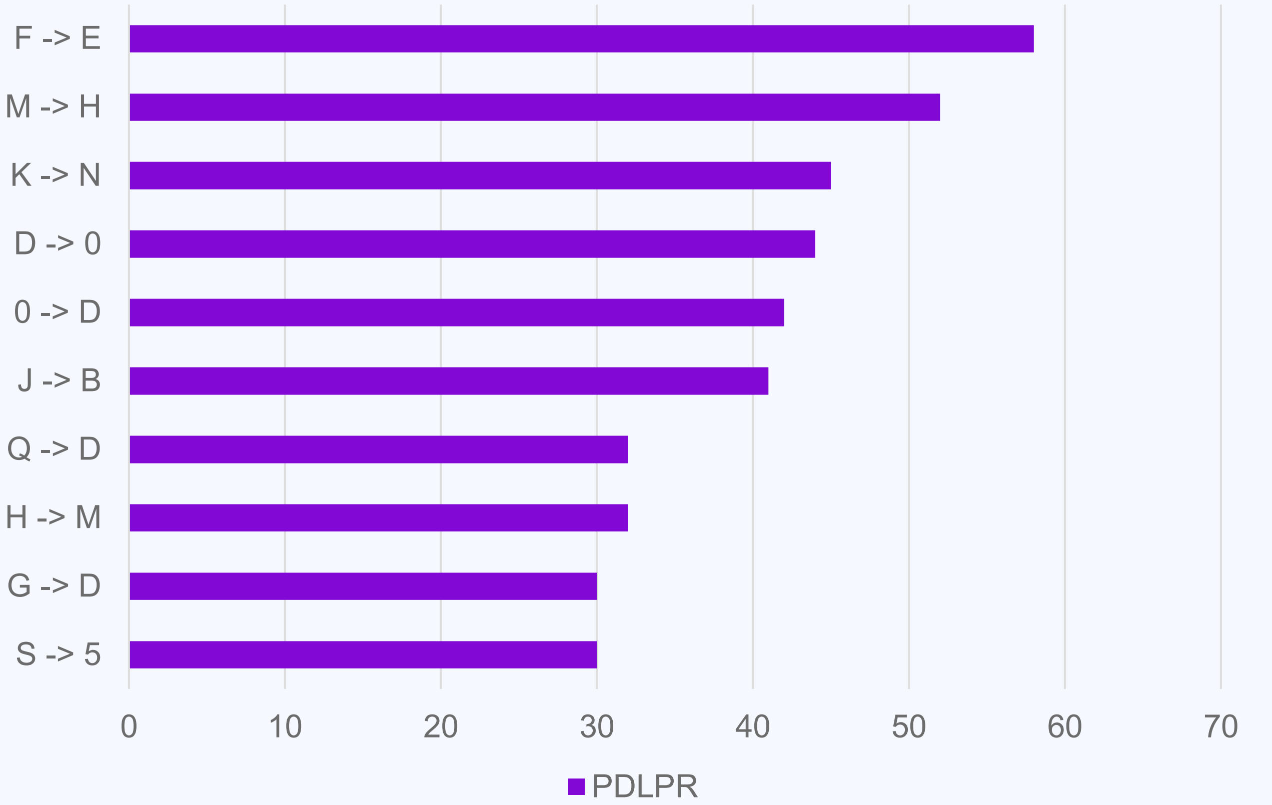
PDLPR results

	base	tilt	blur	rotate	challenge	fn	db	weather	**total**
samples	40000	30216	20611	10053	50003	20967	10132	9999	**150K**
avg loss	0.3216	0.1204	0.1077	0.0700	0.0734	0.0859	0.1089	0.0040	**0.1115**
acc/char	99.45%	88.65%	95.44%	96.14%	94.64%	93.94%	93.28%	99.60%	**95.14%**
acc/plate	98.86%	75.39%	79.45%	84.12%	78.49%	77.51%	76.41%	97.40%	**83.03%**

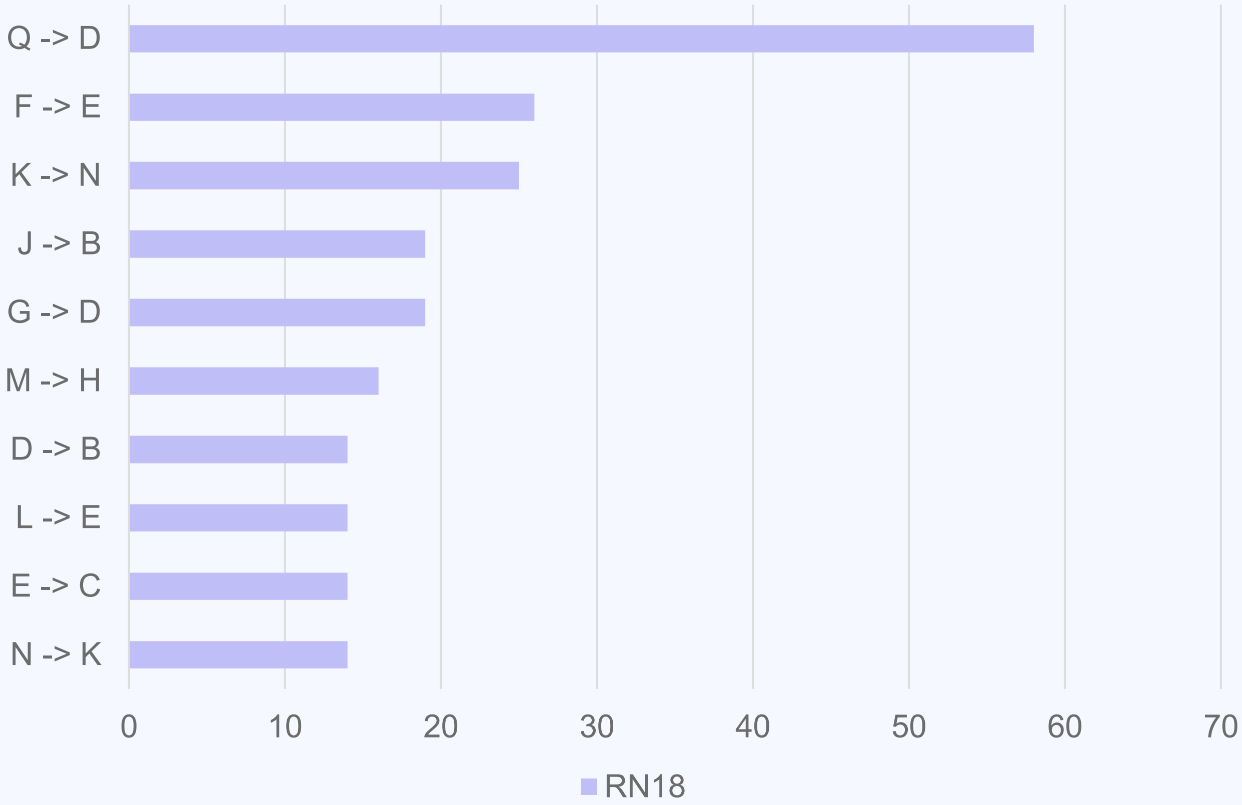
RN18 OCR results

	base	tilt	blur	rotate	challenge	fn	db	weather	**total**
samples	40000	30216	20611	10053	50003	20967	10132	9999	**150K**
avg loss	0.1096	0.0922	0.0552	0.1178	0.1206	0.1511	0.0080	0.1096	**0.1115**
acc/char	99.45%	96.74%	97.44%	98.74%	97.11%	96.82%	96.02%	99.78%	**97.76%**
acc/plate	99.07%	82.22%	85.83%	92.86%	85.94%	84.27%	80.04%	98.46%	**89.59%**

Most common letter and number confusions (1)



Most common letter and number confusions (2)



Conclusions

The specialized **YOLOv5 + PDLPR pipeline** delivers superior results, while **ResNet18**, a general-purpose model, achieves comparable performance with significantly faster training and easier management.

ResNet18 OCR and PDLPR show **comparable performance**, as ResNet18 demonstrates superior character recognition capabilities. However, ResNet18 falls slightly behind in bounding box detection, though it still delivers absolutely acceptable results that meet project requirements.

Both approaches effectively deliver the desired outcomes while offering distinct advantages in specialization versus versatility.

Future works

It would be interesting to evaluate the model proposed in the paper using **YOLOv11**, the latest version of the YOLO architecture, to potentially improve detection performance.

Testing the model on **additional datasets** would help assess its **flexibility** and **generalization capabilities**, especially for different types of license plates across various regions or formats.

Reference

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2. Xu, Z.; Yang, W.; Meng, A.; Lu, N.; Huang, H.; Ying, C.; Huang, L. Towards end-to-end license plate detection and recognition: A large dataset and baseline. In Proceedings of the European Conference on Computer Vision (ECCV), Munich, Germany, 8–14 September 2018.

3. R. K. Prajapati, Y. Bhardwaj, R. K. Jain and D. Kamal Kant Hiran, "A Review Paper on Automatic Number Plate Recognition using Machine Learning : An In-Depth Analysis of Machine Learning Techniques in Automatic Number Plate Recognition: Opportunities and Limitations," 2023 International Conference on Computational Intelligence, Communication Technology and Networking (CICTN), Ghaziabad, India, 2023, pp. 527-532

Thanks for your attention